

Mark schemes

Q1.

Solving $\frac{d^2y}{dt^2} - 6\frac{dy}{dt} + 10y = e^{2t} (*)$

Auxl. Eqn. $m^2 - 6m + 10 = 0$

$(m - 3)^2 + 1 = 0$

PI Completing sq or using quadratic formula to find m.

M1

$m = 3 \pm i$

A1

CF ($y_{CF} =$) $e^{3t}(A \cos t + B \sin t)$

*OE Condone x for t here; ft c's 2 **non-real** values for 'm'.*

M1

For PI try ($y_{PI} =$) ke^{2t}

Condone x for t here

M1

$4k - 12k + 10k = 1 \Rightarrow k = \frac{1}{2}$

A1

GS of (*) is ($y_{GS} =$) $e^{3t}(A \cos t + B \sin t) + \frac{1}{2}e^{2t}$

CF + PI with 2 arb. constants and both CF and PI functions of t only

B1F

[6]

Q2.

(a) PI: $y_{PI} = p + qxe^{-2x}$

$y'_{PI} = qe^{-2x} - 2qxe^{-2x}$

Product rule used

M1

$y''_{PI} = -4qe^{-2x} + 4qxe^{-2x}$

$-4q e^{-2x} + 4qxe^{-2x} + qe^{-2x} - 2qxe^{-2x}$

$$-2p - 2qxe^{-2x} = 4 - 9e^{-2x}$$

Subst. into DE

M1

$$-3q = -9 \quad \text{and} \quad -2p = 4$$

Equating coefficients

m1

$$-3q = -9 \text{ so } q = 3 ;$$

A1

$$-2p = 4 \text{ so } p = -2 ;$$

$$[y_{PI} = 3x e^{-2x} - 2]$$

B1

5

(b) Aux. eqn. $m^2 + m - 2 = 0$

$$(m - 1)(m + 2) = 0$$

Factorising or using quadratic formula OE

PI by correct two values of 'm' seen/used

M1

$$y_{CF} = Ae^x + Be^{-2x}$$

A1

$$y_{GS} = Ae^x + Be^{-2x} + 3x e^{-2x} - 2$$

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(y_{GS}) = c's CF + c's PI, provided 2 arbitrary constants

B1F

3

(c) $x = 0, y = 4 \Rightarrow 4 = A + B - 2$

Only fit if exponentials in GS

B1F

$$\frac{dy}{dx} = Ae^x - 2Be^{-2x} + 3e^{-2x} - 6xe^{-2x}$$

$$\text{As } x \rightarrow \infty, (e^{-2x} \rightarrow 0 \text{ and}) xe^{-2x} \rightarrow 0$$

E1

$$\text{As } x \rightarrow \infty, \frac{dy}{dx} \rightarrow 0 \quad \text{so } A = 0$$

B1

$$\text{When } A = 0, 4 = 0 + B - 2 \Rightarrow B = 6$$

$$y = 6e^{-2x} + 3xe^{-2x} - 2$$

$$y = 6e^{-2x} + 3xe^{-2x} - 2 \quad \text{OE}$$

B1

4

[12]

Q3.

(a) $m\ddot{x} = -4mn^2x - 2mk\dot{x}$

M1A1

$$\ddot{x} + 2k\dot{x} + 4n^2x = 0$$

AG

A1

3

(b) (i) $k = n$

$$p^2 + 2np + 4n^2 = 0$$

$$(p + n)^2 + 3n^2 = 0$$

M1

$$p = -n \pm n\sqrt{3}i$$

A1

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$$x = e^{-nt} (A \cos \sqrt{3}nt + B \sin \sqrt{3}nt)$$

ft provided in correct form

A1F

$$t = 0, x = a \Rightarrow a = A$$

A1

$$\dot{x} = e^{-nt} (-\sqrt{3}nA \sin \sqrt{3}nt + \sqrt{3}nB \cos \sqrt{3}nt)$$

$$-ne^{-nt} (A \cos \sqrt{3}nt + B \sin \sqrt{3}nt)$$

m1

$$t = 0, \dot{x} = 0 \Rightarrow 0 = \sqrt{3}nB - nA$$

$$B = \frac{a}{\sqrt{3}} \left(= \frac{\sqrt{3}a}{3} \right)$$

A1F

$$x = e^{-nt} \left(a \cos \sqrt{3}nt + \frac{\sqrt{3}a}{3} \sin \sqrt{3}nt \right)$$

AG

A1F

7

(ii) $x = 0, a \cos \sqrt{3}nt + \frac{\sqrt{3}}{3} a \sin \sqrt{3}nt = 0$

Condone verification with $t > 0$

M1

$$(e^{-nt} \neq 0)$$

$$\tan \sqrt{3}nt = -\sqrt{3}$$

AG

M1A1

3

(c) (i) $k = 2n$

$$p^2 + 4np + 4n^2 = 0$$

M1

$$(p + 2n)^2 = 0$$

M1

$$x = e^{-2nt}(A + Bt)$$

A1

3

(ii) critical damping

A1F

1

[17]

Q4.

(a) $x = A \cos 4t + B \sin 4t$

$$\dot{x} = -4A \sin 4t + 4B \cos 4t$$

B1

$$\ddot{x} = -16A \cos 4t - 16B \sin 4t$$

B1

$$\text{Alt: } m^2 + 16 = 0$$

B1

$$m = \pm 4i$$

B1

$$x = A \cos 4t + B \sin 4t \quad \text{M1A1}$$

Substitute into $\ddot{x} + 16x = 0$

M1

Satisfactory conclusion

A1

4

(b) $t = 0, \dot{x} = 0: 0 = 0 + 4B \rightarrow B = 0$

AG

B1

1

(c) $x = A \cos 4t$

$$t = \frac{\pi}{12}, x = h: h = A \cos \frac{\pi}{3}$$

M1

$$A = 2h$$

A1

2

(d) $\dot{x} = -8h \sin 4t$

$$\text{Max speed} = 8h \text{ m s}^{-1}$$

B1

1

(e) $F = m(-32h \cos 4t)$

M1

$$|F_{\max}| = 32hm \text{ N}$$

A1

2

[10]

Q5.

- (a) Natural length of AP is $4a$ and natural length of BP is $2a$
When particle is x from equilibrium position:

Tension in AP is $\frac{4mn^2a(2a+x)}{4a}$

M1A1

Tension in BP is $\frac{4mn^2a(a-x)}{2a}$

M1A1

In general position, using $F = ma$:

$$m \frac{d^2x}{dt^2} = \frac{4mn^2(a-x)}{2a} - \frac{4mn^2a(2a+x)}{4a} - 2mn \frac{dx}{dt}$$

M1A1

$$m\ddot{x} = 2mn^2a - 2mn^2x - 2mn^2a - mn^2x - 2mn\dot{x}$$

$$\frac{d^2x}{dt^2} + 2n \frac{dx}{dt} + 3n^2x = 0$$

AG

A1

7

- (b) [Substituting $x = Ae^{pt}$]
 $p^2 + 2p + 3 = 0$

M1A1

$$p = -1 \pm \sqrt{2}i$$

A1

General solution is:

$$x = e^{-t}(A \cos \sqrt{2}t + B \sin \sqrt{2}t)$$

A1

When $t = 0$, $x = \frac{1}{2}a \Rightarrow \frac{1}{2}a = A$

B1

Differentiating:

$$\frac{dx}{dt} = -e^{-t}(A \cos \sqrt{2}t + B \sin \sqrt{2}t) + e^{-t}(-A\sqrt{2} \sin \sqrt{2}t + B\sqrt{2} \cos \sqrt{2}t)$$

M1A1

When $t = 0$, $\frac{dx}{dt} = 0$

$$\Rightarrow 0 = -A + \sqrt{2}B$$

$$A = \frac{1}{2}a, B = \frac{1}{2\sqrt{2}}a$$

A1

$$x = ae^{-t} \left(\frac{1}{2} \cos \sqrt{2}t + \frac{1}{2\sqrt{2}} \sin \sqrt{2}t \right)$$

8

[15]

Q6.

(a) $2 \times 9.8 = \frac{\lambda}{0.5} \times 0.2$

Equilibrium considered to form equation in λ

M1

$$\lambda = \frac{9.8}{0.2} = 49 \text{ N}$$

Correct equation

A1

Correct λ

A1

3

(b) $T = \frac{49x}{0.5} + 2g$

Equation for tension with two terms

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M1

Correct equation

A1

$$2 \frac{d^2 x}{dt^2} = 2g - (98x + 2g)$$

Use of $F = m \frac{d^2 x}{dt^2}$

M1

$$\frac{d^2 x}{dt^2} = -\frac{98}{2}x = -49x$$

Correct equation

A1

ag Correct result from correct working

A1

5

(c) $\text{Period} = \frac{2\pi}{\sqrt{49}}$
Finding period

M1

Correct period

A1

$$t = \frac{1}{4} \times \frac{2\pi}{7} = \frac{\pi}{14} = 0.224 \text{ seconds}$$

Dividing period by 4

M1

Correct time

A1

4

[12]

Q7.

(a) $mg = \frac{\lambda}{l} \times \frac{l}{4}$
Consideration of forces in equilibrium

M1

$$\lambda = 4mg$$

ag Correct λ from correct working

A1

2

(b) (i) $m \frac{d^2x}{dt^2} = mg - T$
 $= mg - \frac{4mg}{l} \left(x + \frac{l}{4} \right)$
Two term expression for T

M1

Correct expression for T

A1

Three term equation of motion

M1

$$= -\frac{4mg}{l}x$$

Simplifying

m1

$$\frac{d^2x}{dt^2} = -\frac{4g}{l}x$$

ag Correct result from correct working

A1

5

$$(ii) \text{ Period} = 2\pi\sqrt{\frac{1}{4g}} = \pi\sqrt{\frac{1}{g}}$$

Identifying ω

Correct period

M1

A1

2

[9]

Q8.

$$(a) \quad T_{AP} = \frac{\lambda \cdot 3\alpha}{4\alpha} = \frac{3}{4}\lambda$$

$$T_{PB} = 4mg \cdot \frac{\alpha}{2\alpha} = 2mg$$

Either

B1

Using $F = ma$ vertically

M1

$$mg + T_{PB} = T_{AP}$$

$$\therefore mg + 2mg = \frac{3}{4}\lambda$$

A1

$$\lambda = 4mg$$

A1

4

- (b) (i) When particle is moved a distance x below the equilibrium position, forces acting on it are

$$mg, T_{AP} = \frac{\lambda(3a+x)}{4a} = \frac{mg(3a+x)}{a},$$

M1

$$T_{PB} = 4mg \cdot \frac{(a-x)}{2a} = \frac{2mg}{a}(a-x)$$

and resistance $\frac{1}{3}mk\dot{x}$

All four forces

M1

[forces 2 and 4 are upwards]

Using $F = ma$ vertically downwards

$$m\ddot{x} = mg + T_{PB} - T_{AP} - \frac{1}{3}mk\dot{x}$$

Dependent on both M1 above

m1

$$m\ddot{x} = mg + \frac{2mg}{a}(a-x) - \frac{mg(3a+x)}{a} - \frac{1}{3}mk\dot{x}$$

A1

$$\ddot{x} - g - \frac{2g}{a}(a-x) + \frac{g(3a+x)}{a} + \frac{1}{3}k\dot{x} = 0$$

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$$\ddot{x} + \frac{1}{3}k\dot{x} + \frac{3gx}{a} = 0$$

A1

$$10 \frac{d^2x}{dt^2} + 2k \frac{dx}{dt} + 5k^2x = 0$$

A1

6

- (ii) Substituting $x = Ae^{nt}$,

$$10n^2 + 2kn + 5k^2 = 0$$

$$n = \frac{-2k \pm \sqrt{4k^2 - 200k^2}}{20}$$

M1

$$= \frac{1}{10}(-k \pm 7ki)$$

A1

$$x = e^{-\frac{k}{10}t} (A \cos \frac{7}{10}kt + B \sin \frac{7}{10}kt)$$

M1 A1ft

$$\text{When } t = 0, x = \frac{a}{2}, A = \frac{a}{2}$$

B1

Differentiating

$$\begin{aligned} \frac{dx}{dt} &= -\frac{k}{10} e^{-\frac{k}{10}t} (A \cos \frac{7}{10}kt + B \sin \frac{7}{10}kt) + \\ &e^{-\frac{k}{10}t} (-\frac{7}{10}kA \sin \frac{7}{10}kt + \frac{7}{10}kB \cos \frac{7}{10}kt) \end{aligned}$$

M1 A1ft

$$\text{When } t = 0, \frac{dx}{dt} = 0, 0 = \frac{k}{10}A + \frac{7}{10}kB$$

M1

$$B = \frac{a}{14}$$

$$x = \frac{a}{14} e^{-\frac{k}{10}t} (7 \cos \frac{7}{10}kt + \sin \frac{7}{10}kt)$$

A1

9

[19]

Q9.

$$(a) \quad mg = \frac{\lambda e}{l}$$

Equation for equilibrium

M1

$$e = \frac{mgl}{\lambda}$$

Correct extension

A1

$$(b) \quad T = -\frac{\lambda}{1} \left(x + \frac{mgl}{\lambda} \right)$$

Expression for tension

Correct expression

A1

$$m \frac{d^2 x}{dt^2} = mg - T$$

Application of Newton's 2nd law with T and mg

m1

$$= mg - \frac{\lambda}{1} \left(x + \frac{mgl}{\lambda} \right)$$

Simplified to correct final answer

A1

5

$$(c) \quad \omega = 2\pi \sqrt{\frac{\lambda}{ml}}$$

$$P = 2\pi \sqrt{\frac{ml}{\lambda}}$$

Finding period

M1

Correct final answer

A1

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2

(d) Increased by a factor of 2

Increased

B1

Factor of 2

B1

2

[11]

Q10.

$$(a) \quad t = \frac{1}{4} \times \frac{2\pi}{5}$$

$$\omega = 5$$

$\frac{1}{4}$ of their period

B1 M1

$$= \frac{\pi}{10}$$

A1

3

(b) $v_{\max} = 5a$

$$v_{\max} = 5a$$

B1

$$\left(\frac{5a}{2}\right)^2 = 5^2 (a^2 - x^2)$$

Use of $v^2 = \omega^2(a^2 - x^2)$ with $v = \frac{5a}{2}$

M1

$$x^2 = \frac{3a^2}{4}$$

Correct equation
Solving for x

A1 m1

$$x = \frac{a\sqrt{3}}{2}$$

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Answer as given or equivalent

A1

5

[8]

Q11.

(a) Substituting $z = Ae^{nt}$,

$$2n^2 + 6n + 9 = 0$$

M1

$$n = \left(-6 \pm \frac{\sqrt{36 - 72}}{4} \right)$$

$$n = -\frac{3}{2} \pm \frac{3}{2}i$$

A1

$$\therefore x = e^{-\frac{3}{2}t} \left(A \cos \frac{3}{2}t + B \sin \frac{3}{2}t \right)$$

M1 A1ft

When $t = 0$, $x = a$; $a = A$

B1

Differentiating

$$\frac{dx}{dt} = -\frac{3}{2} e^{-\frac{3}{2}t} \left(A \cos \frac{3}{2}t + B \sin \frac{3}{2}t \right) + \frac{3}{2} e^{-\frac{3}{2}t} \left(-A \sin \frac{3}{2}t + B \cos \frac{3}{2}t \right)$$

Use of product formula

M1 A1

When $t = 0$, $\frac{dx}{dt} = 0$; $0 = -\frac{3}{2}A + \frac{3}{2}B$

M1

$$B = a$$

$$\therefore x = a e^{-\frac{3}{2}t} \left(\cos \frac{3}{2}t + \sin \frac{3}{2}t \right)$$

A1

9

- (b) Damping motion is light damping
And one reason

B1

since the particle oscillates, with its amplitude reducing to zero

B1

2

Or

$e^{-kt} \times \text{trig. term}$

Or

Sketch from graphics calculator

Full reason

[11]

Examiner reports

Q1.

The method to find the general solution of this second order differential equation was well understood by almost all students. The most common error was to incorrectly solve the auxiliary equation to obtain roots $3 \pm 2i$ although many then presented correct follow through work to score 5 of the 6 marks.

Q2.

Most candidates were able to find the correct values of the constants p and q so that the given expression was a particular integral of the differential equation. An even higher proportion of candidates scored full marks for their general solution in part (b) as examiners allowed follow through on incorrect values for p and q . However, less than 20% of the candidates scored full marks for their answers to part (c). This was almost entirely due to no explicit consideration being given to the value of xe^{-2x} as $x \rightarrow \infty$. More surprisingly, a significant minority of candidates could not correctly deal with the boundary

condition $\frac{dy}{dx} \rightarrow 0$ as $x \rightarrow \infty$ as they had an e^x term in their final answer.

Q3.

This question was answered well in most cases. Part (a) was popular and mostly yielded full marks. Solutions to the auxiliary equation in part (b)(i) were often fully correct, with the most frequent error being the omission of 'i' in the solution, despite the obvious nature of the roots. Some chose a verification method for part (b)(ii) but only consideration of a general solution was sufficient for full credit. Solutions in part (c) were impressive.

Q4.

Again, this question was answered well, with candidates showing good knowledge and understanding of this topic. The appropriate techniques were usually applied efficiently to produce correct solutions.

Q5.

Weaker candidates created the printed differential equation in part (a) by dubious means. It was necessary to find the equilibrium position and then use the extensions to be $2a + x$ in string AP and $a - x$ in string BP . Part (b) was answered very well by most candidates.

The only common error was in trying to insert the boundary conditions and finding $\frac{dx}{dt}$, when $\frac{d}{dt} \cos \sqrt{2}t$ sometimes became $-\sin \sqrt{2}t$ rather than $-\sqrt{2} \sin \sqrt{2}t$.

Q6.

The candidates did well with part (a) and for quite a number of candidates these were the only marks that they obtained on this question.

In part (b), the majority of the candidates totally ignored the weight of the sphere and produced invalid arguments. There were some candidates who could produce the

required result without difficulty.

In the last part, there were quite a few candidates who found the period, but did not realise that the time needed was a quarter of the period.

Q7.

Part (a) of this question was done very well by almost all of the candidates. Part (b)(i) was very much more demanding and there were very few correct solutions. Most of the candidates did not find a two term expression for the tension in the string and did not include gravity when applying Newton's second law. In contrast, part (b)(ii) was done quite well, although some candidates did have difficulty simplifying their results.

Q8.

Most candidates answered part (a) correctly but a number did not use $mg + T_{PB} = T_{AP}$ and fiddled the algebra to give the printed result, for which they gained no marks.

In part (b), candidates had to find tensions in the two springs AP and PB , with extensions $(3a + x)$ and $(a - x)$ respectively, and then use $F = ma$ downwards, giving

$m\ddot{x} = mg + T_{PB} - T_{AP} - \frac{1}{5}mkx$. Without these equations candidates were not given credit for producing the printed differential equation.

All candidates used the printed equation given in part (b)(i) to find x in part (ii). A number of candidates lost k in a variety of places, but virtually all obtained a result similar to that required.

Q9.

There were many good answers to part (a). Virtually all those who realised that they needed to consider when the forces were in equilibrium gained both marks. Part (b) was found to be very much more difficult. Several candidates did not include the weight of the particle at any stage of their solutions. Some others produced very contrived arguments to produce the negative sign in the differential equation. Part (c) was often done well, and those who did not gain marks on this question did not seem to know how to proceed. In part (d), several of the candidates stated that the period would increase, but a smaller number were able to quantify the increase correctly.

Q10.

There were some good solutions to this question, but many candidates found it difficult to make much progress. Part (a) in particular caused a lot of difficulty. Quite a number of candidates did not realise that a quarter of the period was needed, often using a half instead. In part (b), there were more correct solutions, but some candidates did not seem to know how to approach the problem.

Q11.

Part (a) was completed well by most candidates. However the use of the formula to solve the quadratic equation $2n^2 + 6n + 9 = 0$ was completed incorrectly by a number of candidates. Most appreciated that the damping was light, although the reason given for this was often imprecise.